1. Single choice (10%, 2% for each)

(1) Which choice ( ) is the task of digit image mid-level process.

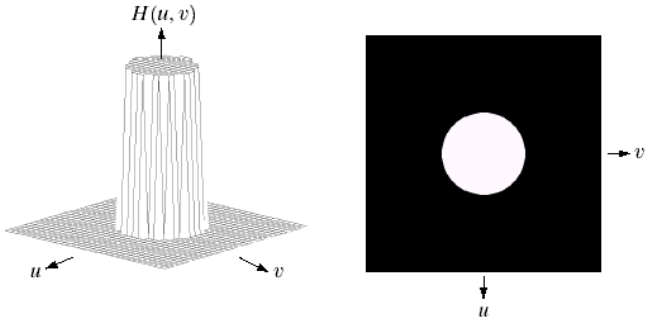
A. Image understanding B. Image sharpening

C. Image segmentation D. Image smoothing

(2) The storage of a 256*512 gray image with 256 intensity levels is ( ) bytes.

A. 128KB B. 1M C. 0.5M D.256KB

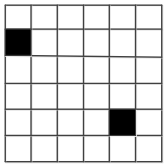
(3) Which filter ( ) has the transfer function shown as below?



A. Gaussion lowpass filter B. Ideal lowpass filter

C. Butterworth highpass filter D. Ideal highpass filter

(4) Considering the following figure, compute the D_2 distance (called the Euchildean distance)between two black pixels: ( )



A.3 B. 4 C. 5 D.6

(5) Which color model ( ) is utilized in color monitors.

A. RGB B. CMY

C. HSI D. All of them

2.True or False (10%, 2% for each)

(1) Image compression is an objective process and its results can be quantitative.( )

(2) The chain-coded of boundary does not depends on the starting point. ( )

(3) In the frequency domain, we can use lowpass filter to sharpen the image. ( )

(4) In the frequency domain, the function of highpass filter is to smooth the image. ( )

(5) The objects of image segmentation are not intersected. ( )

3. (10%) If we apply nearest neighbor interpolation and bilinear interpolation respectively, what would be the values of the new pixels at points P1 and P2? (both methods).

	3		P1	9
	P2			
2				10

4. (10%) An image with 6 gray-levels as follow:

1	3	2	1	3
1	0	0	0	5
4	3	0	4	5
4	2	1	5	3
1	3	2	5	3

(1) Plot the histogram of the image.

(2) Process the histogram equalization.

(3) Give the output image after histogram equalization.

5.(8%) Prove the following translation property.

$$f(x,y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u-u_0,v-v_0)$$

6.(8%) Give an intensity transformation function for spread the intensities of an image so the range from [0:20] to [0:40], from[20:50] to [40:80], from[50:100] to [80:100] and plot the function.

7.(10%) The image is as following:

(1) Compute the results using average(3*3 mask), Midpoint and Median filter to the image shown as follow(Do not process the border of the image).

4	12	2	6
10	6	10	8
6	8	4	6
6	12	4	10

Original image

4	12	2	6
10			8
6			6
6	12	4	10

The results of Average filter

4	12	2	6
10			8
6			6
6	12	4	10

The results of Midpoint filter

4	12	2	6
10			8
6			6
6	12	4	10

The results of Median filter

8.(12%) The image is as following:

3	5	3	2	5
5	3	1	5	3
3	1	3	2	4
1	5	3	5	1

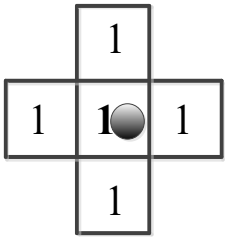
(1)Compute the huffman coding of every pixels.(9%)

(2)compute the compression ratio and relative data redundancy, suppose the original code is 8 bit. (Round to 2 decimal places) (3%)

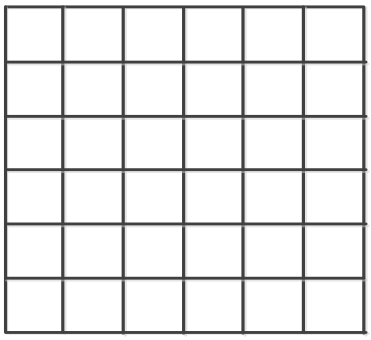
9.(10%) Opening is an operation of the morphological image processing,

Apply the Opening processing to the following image.(two steps)

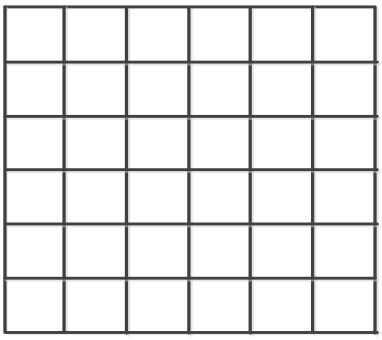
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	1	0
1	1	1	1	1	0
0	1	1	1	1	1
0	0	0	1	0	0



(a)Original image (b)Structure Element (Black dot denotes origin)



Step 1



Step 2

10.(6%)The following code implements the multiple thresholding algorithm to segment the grayscale image into three classes.

(a) Explain the mean of line 2.

(b) Explain the mean of line 3.

(c) Explain the mean of line 9.

```
1   imgdata=imread('cameraman.tif');
2   [rows,cols]=size(imgdata)
3   segresult=zeros([rows,cols],'uint8');
4   for row=1:rows
5         for col=1:cols
6               if imgdata(row,col)<100
7                     segresult(row,col)=1;
8               elseif imgdata(row,col)<180
9                     segresult(row,col)=2;
10              else
11                    segresult(row,col)=3;
12              end
13        end
14  end
```

11.(6%)Design a Basketball Recognition Systems. Draw the principle diagram. Explain the function and method used in each diagram.